一些常见结论

Contents

1 Abstract Algebra 抽象代数

Proposition 1.1. *Let S be a monoid. If an element $a \in S$ satisfies that there exists $b, c \in S$ s.t. $ba = 1, ac = 1$, then $b = c$.*

Proposition 1.2. *Let R be an integral domain. Then $R = \cap_\alpha R_{\mathfrak{m}_\alpha}$, where $\mathfrak{m}_\alpha$ varies over all the maximal ideals of R.*

Theorem 1.1 (***Schur' Theorem***)**.** *Let G be a group. If $Z(G)$ is of finite index in G, then $G' = [G, G]$ is finite.*

Proposition 1.3. *G is a infinite group. If every nontrivial proper subgroup of G is of finite index, then G is cyclic.*

Proposition 1.4. *G is a finite group, $H \leq G$ subgroup. Show there is a complete set of left cosets being a complete set of right cosets at the same time.*

Proposition 1.5. *All groups of order pq are either isomorphic to $\mathbb{Z}_{pq}$ (abelian case) or $\mathbb{Z}_p \rtimes \mathbb{Z}_q$ (nonabelian case).*

Proposition 1.6. *G is a p-group. Then all maximal subgroups of G is normal and of index p.*

Proposition 1.7. *G is a finite group. Assume that $v_p(G) = n$. Then the number of subgroups of order p^i, n_1, satisfies that $n_1 \equiv 1 \pmod p$, for $\forall 1 \leq i \leq n$.*

Proposition 1.8. *G is a nonabelian simple group of order 60. Then $G \cong A_5$.*

Proposition 1.9. *G is a finite group, $N \trianglerighteq G$ a normal group, $p \mid \sharp(N)$. Assume that P is a Sylow p-group of N. Then $G = N_G(P)N$.*

2 Topology 拓扑

2.1 Basic topology

Proposition 2.1. *Let X be a topological space., $S \subseteq X$ subset. Then S is closed in X if and only if X can be covered by open subsets U_α s.t. $S \cap U_\alpha$ is closed in U_α.*

2.2 Irreducibility

Proposition 2.2. *Let X be an irreducible topological space. Then for all nonempty open subset $U \subseteq X$, U is dense.*

Proposition 2.3. *Let X be a topological space. Then every maximal irreducible subspace of X is closed.*

Remark 2.1. *For convenience, we call maximal irreducible subspace irreducible component.*

Proposition 2.4. *Let X be a noetherian topological space. Then X has only finitely many irreducible components.*

2.3 Dimension theory

3 Commutative Algebra 交换代数

In this section, all rings are commutative.

3.1 Radical

Proposition 3.1. *Let R be a ring. The nilradical $\mathfrak{N}$ is the intersection of all prime ideals of R.*

Proposition 3.2. *Let R be a ring. Define the Jacobson radical $\mathfrak{R}$ is the intersection of all maximal ideals of R. We have $x \in \mathfrak{R} \iff 1 - xy$ is a unit in R for all $y \in R$.*

Proposition 3.3. *Let R be a ring, $I, J \subseteq R$ ideals. If $\sqrt{I}$ and $\sqrt{J}$ are coprime, then I and J are coprime.*

Proposition 3.4. *Let R be a ring. Then every idempotent of $R/\mathfrak{N}$ lifts to some idempotent of R. In fact, the lifting is unique.*

3.2 Localization

Proposition 3.5. *Let R be a ring. Then $R = \cap_{\mathfrak{m} \in Max(R)} R_{\mathfrak{m}}$.*

Proposition 3.6. *Let R be a reduced ring. Then $S^{-1}R$ is reduced for all multiplicatively closed subset S.*

Corollary 3.1. *Let R be a reduced ring. Then $R_{\mathfrak{p}}$ is a field for all minimal prime ideal $\mathfrak{p}$.*

3.3 Modules

Proposition 3.7. *Let R be a noetherian ring. Then submodule of finitely generated R-module is still finitely generated.*

Proposition 3.8. *Let R be a noetherian ring. Then for every finite R-module M, there is a filtration*

$$0 = M_0 \subset M_1 \subset \cdots \subset M_n = M \tag{1}$$

such that $M_i/M_{i-1} \cong A/\mathfrak{p}_i$ for some prime ideal $\mathfrak{p}_i$.

Corollary 3.2. *Let R be a noetherian ring. There is a filtration*

$$0 = I_0 \subset I_1 \subset \cdots \subset I_n = R \tag{2}$$

such that $I_i/I_{i-1} \cong A/\mathfrak{p}_i$ for some prime ideal $\mathfrak{p}_i$. And for all minimal prime ideal $\mathfrak{q}$ of R, $\mathfrak{q}$ appears exactly $\operatorname{length} A_{\mathfrak{q}}$ *times.*

3.4 Algebras

⋆ Finiteness

Proposition 3.9. *Let B be a finite A-algebra with homomorphism $\varphi : A \to B$. Assume $\mathfrak{m}$ is a maximal ideal in A. Then $B_{\mathfrak{m}} \cong \oplus_i B_{\mathfrak{p}_i}$, where $\mathfrak{p}_i$ varies over all prime ideals of B lying over $\mathfrak{m}$.*

⋆ Integrality

Proposition 3.10. *Let L/K be an algebraic field extension, $A \subseteq B$ and $\operatorname{Frac} A = K$. Assume B is the integral closure of A in L. Then $\operatorname{Frac} B = L$.*

Proposition 3.11. *Let L/K be an algebraic field extension, $A \subseteq B$ and $\operatorname{Frac} A = K$. Assume B is the integral closure of A in L. Then $B \otimes_A K = L$.*

Proposition 3.12. *Let $A \subseteq B$ be an integral extension with A integral domain. Then A is a field if and only if B is a field.*

⋆ Flatness

Proposition 3.13. *Let $f : A \to B$ be a ring homomorphism. Then f is flat if and only if for all $\mathfrak{P} \in \operatorname{Spec} B$ lying over some $\mathfrak{p} \in \operatorname{Spec} A$, the induced map $A_{\mathfrak{p}} \to B_{\mathfrak{P}}$ is flat.*

Proposition 3.14. *Let $f : A \to B$ be an etale homomorphism i.e. flat amd unramified. If A is regular, then B is regular.*

Proposition 3.15. *Let $f : A \to B$ be a homomorphism between local rings. Then f is flat if and only if f is faithfully flat.*

⋆ Faithful Flatness

Proposition 3.16. *Let $f : A \to B$ be a faithfully flat homomorphism, $M \in \mathbf{Mod}_A$. Then $M = 0$ if and only if $M \otimes_A B = 0$.*

Corollary 3.3. *Let $f : A \to B$ be a faithfully flat homomorphism. Then for each ideal $I \subseteq A$, we have $IB \cap A = I$.*

Corollary 3.4 (***Faithfully Flat Descent***)**.** *Let $f : A \to B$ be a faithfully flat homomorphism. Then for all homomorphism $\varphi : M \to N$ between A-modules, φ is an isomorphism if and only if $\varphi \otimes_A B$ is an isomorphism.*

3.5 Dimension theory

Proposition 3.17. *Let R be an artinian integral domain. Then R is a field.*

3.6 Local rings

Proposition 3.18. *Let R be a local ring, $\mathfrak{m} \subseteq R$ the maximal ideal. If $\dim_{R/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) = n$, then $\mathfrak{m}$ is generated by exactly n elements.*

Corollary 3.5. *Let R be a local ring, $\mathfrak{m} \subseteq R$ the maximal ideal. If $\dim_{R/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) = 1$, then $\mathfrak{m}$ is principal.*

Proposition 3.19. *Let $(R, \mathfrak{m})$ be a local ring with residue field k, M finitely generated R-module. Then the following conditions are equivalent:*
(1)M is free
(2)M is projective
(3)M is flat
(4)$\mathrm{Tor}_1^R(M, k) = 0$.

Corollary 3.6. *Let R be a DVR, M finite over R. Then M is flat over R if and only if it is torsion-free over R.*

⋆ Valuation Rings

Proposition 3.20. *Let R be a valuation ring, $a, b \in R$. Then the ultrametric inequality $v(a+b) \geq \min\{v(a), v(b)\}$ takes equality whenever $v(a) \neq v(b)$.*

⋆ Henselian Local Rings

Proposition 3.21. *Let R be a local ring. Then R is Henselian if and only if any finite R-algebra is product of local rings.*

Corollary 3.7. *Let R be a Henselian local ring, S finite R-algebra. If S is an integral domain, then S is a local ring.*

Proposition 3.22. *Let R be UFD. Then R is integrally closed.*

Proposition 3.23. *Let $B \in Alg_A$ with ring homomorphism $f : A \longrightarrow B$. For $\mathfrak{p} \in \operatorname{Spec} A$, if $\mathfrak{p}^{ec} = \mathfrak{p}$, then $\mathfrak{p} \in im(f^*)$.*

Proposition 3.24. *Let A be a graded ring and M, N, L graded A-modules. Then* $\mathrm{Hom}(M \otimes_A N, L) \cong \mathrm{Hom}(M, \mathrm{Hom}(N, L))$.

Proposition 3.25. *Let $\varphi : A \longrightarrow B$ be a homomorphism between noetherian rings. Then for each minimal prime ideal $\mathfrak{p}$ containing $\ker(\varphi)$ such that $\varphi(\mathfrak{p}B) \neq B$, we can find $\mathfrak{P} \in \operatorname{Spec} B$ lying over $\mathfrak{p}$.*

Corollary 3.8. *Let $\varphi : A \longrightarrow B$ be a homomorphism between noetherian rings. Then for each prime ideal $\mathfrak{p}$ containing $\ker(\varphi)$ such that $\varphi(\mathfrak{p}B) \neq B$, we can find $\mathfrak{P} \in \operatorname{Spec} B$ lying over $\mathfrak{p}$.*

4 Linear Algebra 线性代数

Proposition 4.1. *If U,V are subspaces of dimension r,s of a vector space of W of dimension n, then* $U \cap V$ *is a subspace of dimension more than r + s - n.*

5 Representation Theory 表示论

Theorem 5.1 (***Bornside*** $p^a q^b$ ***Theorem***)**.** *G is a finite group of order* $p^a q^b$*, where* p, q *are prime numbers. Then G is solvable group.*

Theorem 5.2 (***Clifford Theorem***)**.** *G is a finite group,* $N \trianglerighteq G$ *a normal subgroup. Assume that* $\chi \in Irr(G)$ *and* $\varphi \in Irr(N)$ *satisfy that* φ *is a direct summand of* $Res_N^G(\chi)$*. Then* $Res_N^G(\chi) = e\sum_{i=1}^n \varphi^{g_i}$*, where* $e \in \mathbb{N}$*,* φ^{g_i} *is given by* $x \longmapsto \varphi(g_i x g_i^{-1})$ *and* $g_1, \cdots, g_n$ *is a transversal of N in G.*

6 Algebraic Geometry 代数几何

6.1 Classic examples

Proposition 6.1. *Let* $\mathcal{L}$ *be an invertible sheaf on projective line* $\mathbb{P}_k^1$*. Then* $\mathcal{L}$ *must be isomorphic to* $\mathcal{O}_{\mathbb{P}_k^1}(n)$ *for some n.*

6.2 Spectrum

Proposition 6.2. *Let* $\varphi : A \to B$ *be a ring homomorphism. Assume that* $f : \operatorname{Spec} B \to \operatorname{Spec} A$ *is the corresponding morphism of spectrum. Then* $\operatorname{im} f \subseteq V(\ker \varphi)$*.*

Corollary 6.1. *Let* $\varphi : A \to B$ *be a homomorphism between noetherian rings. Assume that* $f : \operatorname{Spec} B \to \operatorname{Spec} A$ *is the corresponding morphism of spectrum. Then* $\operatorname{im} f = \{\mathfrak{p} \in V(\ker \varphi) \big| \varphi(\mathfrak{p})B \neq B\}$*.*

Corollary 6.2. *Let* $\varphi : A \to B$ *be a ring homomorphism. Assume that the corresponding morphism* $f : \operatorname{Spec} B \to \operatorname{Spec} A$ *is surjective. Then* $\ker \varphi \subseteq \mathfrak{N}_A$*.*

Proposition 6.3. *Let* $\varphi : A \to B$ *be a surjective ring homomorphism. Then the corresponding morphism* $f : \operatorname{Spec} B \to \operatorname{Spec} A$ *is injective.*

Proposition 6.4. *Let* $\varphi : A \to B$ *be a ring homomorphism. Assume that* $f : \operatorname{Spec} B \to \operatorname{Spec} A$ *is the corresponding morphism of spectrum. Then f is a closed immersion if and only if* φ *is surjective.*

6.3 Properties of morphisms

$\star$ Separatedness

Proposition 6.5. *Let X be a reduced scheme and Y be a separated scheme. Assume f and g are two morphisms from X to Y, agreeing on an open dense subset of X. Then* $f = g$*.*

Remark 6.1. *With this proposition, the equivalence classes of rational classes from reduced scheme to separated scheme are well behaved. This is why we always assume schemes are reduced and separated when we consider birational maps.*

⋆ **Flatness**

Proposition 6.6. *Let $f : X \to Y$ be a morphism of schemes. Then f is flat if and only if for each $x \in X$, the local ring $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$.*

Proposition 6.7. *Let $f : X \to Y$ be a flat morphism, Y irreducible with generic point η. Assume the fiber of f at η is irreducible. Then X is also irreducible.*

⋆ **Faithful Flatness**

Proposition 6.8 (***Faithfully Flat Descent***)**.** *Let $f : X \to Y$ be a faithfully flat morphism of schemes. Then for all map $\varphi : \mathcal{F} \to \mathcal{G}$ between quasi-coherent sheaves on Y, φ is an isomorphism if and only if $f^*\varphi$ is an isomorphism.*

Corollary 6.3. *Let $f : X \to Y$ be a faithfully flat morphism of schemes. Then for each morphism $\varphi : T \to S$ between schemes affine over Y, φ is an isomorphism if and only if $\varphi \times_Y X$ is an isomorphism.*

Remark 6.2. *In fact, if moreover assume f is quasi-compact, there are lots of properties of morphisms that can be checked after base change by f, which is called descent property local in the fpqc topology. One can refer to* Stacks Project 02YJ *for more details.*

6.4 Dimension theory

Proposition 6.9. *Let X be an irreducible affine scheme. Then for any nonempty open subset $U \subseteq X$, we have that $\dim U = \dim X$.*

Remark 6.3. *This is not true for general topological space.*

Proposition 6.10. *Let X be a noetherian topological space, $Y \subseteq X$ constructible set. Then $\dim Y = \dim \overline{Y}$.*

Proposition 6.11. *Let X be an irreducible scheme locally of finite type over field k. Then for any field extension K/k, we have $X \times_k K$ is of pure dimension of $\dim X$.*

6.5 Universal injectivity

Proposition 6.12. *Let $f : \operatorname{Spec} L \to \operatorname{Spec} K$ be a morphism of schemes. Then f is universally injective if and only if field extension L/K is purely inseparable.*

Proposition 6.13. *Let $f : X \to Y$ be a morphism of schemes. Then f is universally injective if and only if $\Delta_{X/Y} : X \to X \times_Y X$ is surjective.*

6.6 Quasi-coherent sheaves

Proposition 6.14. *Let X be a locally noetherian scheme, $\mathcal{F}$ coherent sheaf on X. Then* $\operatorname{Supp}\mathcal{F} := \{x \in X \big| \mathcal{F}_x \neq 0\}$ *is a closed subset of X.*

Remark 6.4. *For one single section s, there are two meaning of support of s over X. One is the set $\{x \in X \big| s_x \neq 0\}$. And the other is the set $\{x \in X \big| s(x) = 0\}$. Both of the sets are closed subset of X. But support of a sheaf is not closed in general.*

6.7 Vector bundles

Proposition 6.15. *Let $\mathbb{P}^n_k$ be a projective space. Then there is a canonical inclusion* $\mathcal{O}_{\mathbb{P}^n_k}(-1) \hookrightarrow \mathcal{O}_{\mathbb{P}^n_k}^{\oplus n+1}$.

Corollary 6.4. *Let X be a scheme, $\mathcal{E}$ a vector bundle on X with associated projective bundle $\mathbb{P}(\mathcal{E})$. Then there is a canonical inclusion $\mathcal{O}_{\mathbb{P}(\mathcal{E})}(-1) \hookrightarrow \pi^*\mathcal{E}$, where $\pi : \mathbb{P}(\mathcal{E}) \to X$ is the natural morphism.*

Proposition 6.16. *Let X be a scheme, $\mathcal{E}$ a vector bundle on X with associated projective bundle $\mathbb{P}(\mathcal{E})$. Then $\mathbb{P}(\mathcal{E})$ is irreducible if and only if X is irreducible.*

6.8 Exceptional locus

Proposition 6.17. *Let $f : X \to Y$ be a birational morphism between reduced and separated schemes, $g : Y \dashrightarrow X$ the birational inverse of f. Then $\{x \in X \big| g \text{ is not defined at } f(x)\}$ and $X \setminus U_X$ are the same set, denoted by* $\operatorname{Ex}(f)$, *where U_X denotes the maximal open dense subset of X such that $f\big|_{U_X}$ is an isomorphism.*

Proposition 6.18. *Let $f : X \dashrightarrow Y$ be a birational map between reduced, noetherian and separated schemes, $g : Y \dashrightarrow X$ the birational inverse of f. Then for any prime divisor D on Y not contained in* $\operatorname{Sing} Y$, *the generic point η of D is contained in U_Y.*

6.9 Ramification theory

Proposition 6.19. *Let $f : X \to Y$ be a finite morphism between smooth algebraic curves of degree d. Assume $P_1, \cdots, P_r$ are all preimages of $Q \in Y$. Then*

$$\sum_{i=1}^{r} e_{P_i} = d \tag{3}$$

6.10 Intersection theory

Proposition 6.20. *Let X be a scheme over field k, D Cartier divisor on X. If K/k is a field extension, $X_K := X \times_k K$ with projection $p : X_K \to X$, then for all $Z \in Z_d(X)$,* $\deg_{p^*D} p^*Z = \deg_D Z$.

7 Homological Algebra

Proposition 7.1. *Let $\mathscr{A}$ be an abelian category. Assume $I^\bullet$ be an acyclic bounded-below complex of injective objects. Then $I^\bullet$ is contractible i.e. $\mathrm{id}_{I^\bullet}$ is null-homotopic. Dually, acyclic bounded-above complex of projective objects is contractible.*

8 Proofs

8.1 Abstract Algebra 抽象代数

Proposition 1.1. $b = b1 = bac = 1c = c$ □

Proposition 1.2. If there exists $z \notin R$, while $z \in \cap_\alpha R_{\mathfrak{m}_\alpha}$, consider the ideal $I = \{x \in R | xz \in R\}$. Since $1 \notin I$, we have I is a proper ideal of R, thus there is a maximal ideal $\mathfrak{m}$ s.t. $I \subset \mathfrak{m}$. However, it is obvious that $z \notin R_\mathfrak{m}$, contradiction! □

Proposition 1.1. See in the website, Schur's Theorem on commutator subgroup. □

Proposition 1.3. Take normal cyclic subgroup H s.t. the index of H is minimal. Consider the centralizer C of H, and use Theorem 1.1 to show $C = H$. Finally, suppose there is an element $g \notin H$, try to make contradiction. □

Proposition 1.4. Consider the double coset for H. □

Proposition 1.5. Abelian case by Fundamental Theorem of Finitely Generated Abelian Groups and nonabelian case by Sylow Theorem. □

Proposition 1.6. By induction. □

Proposition 1.7. It suffices to prove the case for p-group. G is a p-group of order p^n, denote the number of maximal subgroups of G by m_G. By induction, we assume that $m_G \equiv 1 \pmod{p}$ for $\forall 1 \leq n \leq k$.

Now consider the $k+1$ case. If $m_G = 1$, then done. Otherwise, assume P, P' are two different maximal subgroups of G. Note that by Proposition 1.6, we have $P \trianglelefteq G$, $P' \trianglelefteq G$ and $PP' = G$. In addition, $PP'/P \cong P'/P \cap P'$. Thus $\sharp(P \cap P') = p^{k-1}$. For each maximal subgroup H of P, P' such that $P' \cap P = H$ one-to-one corresponds to a subgroup of G/H of order p except P/H. The correspondence gives a bijection. Thus $m_G \equiv 1 + \sum_{\text{H maximal subgroup of P}} (m_{G/H} - 1) \equiv 1 \pmod{p}$. □

Proposition 1.8. Obviously, G has no proper subgroup of index less than 5. Similarly, by Sylow's Theorem, the number of Sylow 3-groups and Sylow 5-groups are 10 and 6 respectively. And the number of Sylow 2-groups is 5 or 15. It is obvious to show $G \cong A_5$ in the first case. For the other case, there must exists two different Sylow 2-groups P and Q such that $P \cap Q \neq \{1\}$. Assume that $\langle a \rangle = P \cap Q$, then $o(a) = 2$. Consider the order of $C_G(a)$. □

Proposition 1.9. For any $g \in G$, $g^{-1}Pg \subseteq g^{-1}Ng = N$. Thus $g^{-1}Pg$ is still some Sylow p-group of N. By Sylow's Theorem, there exists $n \in N$ such that $g^{-1}Pg = n^{-1}Pn$. Then $g = (gn^{-1})n$, where $gn^{-1} \in N_G(P)$. □

8.2 Topology 拓扑

8.2.1 Basic topology

Proposition 2.1. "⇒": obviously. "⇐": consider $U_\alpha \setminus S$ □

8.2.2 Irreducibility

Proposition 2.2. Suppose that U is not dense, then there exist $x \in X$ and open neighbourhood $W \ni x$ such that $W \cap U = \varnothing$, contradiciting to irreducibility. Hence U is dense in X. □

Proposition 2.2. Suppose V is a maximal irreducible subspace of X. Consider the closure of V. Suffice to show $\overline{V}$ is closed. □

Proposition 2.4. Assume there is infinitely many irreducible components. Consider the descending chain of closed subsets

$$V_1 \supseteq V_1 \cap V_2 \supseteq \cdots \supseteq V_1 \cap \cdots \cap V_n \supseteq \cdots \tag{4}$$

where V_i are distinct irreducible components. □

8.2.3 dimension theory

8.3 Commutative Algebra 交换代数

8.3.1 Radical

Proposition 3.1. See in the Atiyah. □

Proposition 3.2. See in the Atiyah. □

Proposition 3.3. $r(I \cup J) = r(r(I) \cup r(J)) = r(1) = (1)$ □

Proposition 3.4. Assume $\overline{x}$ is the idempotent in $A/\mathfrak{N}$, we have $x^2 - x$ is nilpotent. Thus $\exists n \in \mathbb{N}_+$ s.t. $x^n(1-x)^n = 0$. As (x) and (1 - x) are coprime, we have r(x) and r(1 - x) are coprime. Note that $r(x^n) = r(x)$ and $r((1-x)^n) = r(1-x)$, by Proposition 3.3, get (x^n) and $((1-x)^n)$ are coprime. By Chinese Remainder Theorem, we have

$$A = A/(0) = A/(x^n(1-x)^n) \cong A/(x^n) \times A/((1-x)^n) \tag{5}$$

Take e be the preimage of $(0, 1) \in A/(x^n) \times A/((1-x)^n)$, e is the lifting of x. □

8.3.2 Localization

Proposition 3.5. Consider natural map $f : R \longrightarrow \cap_{\mathfrak{m}\in Max(R)} R_{\mathfrak{m}}$. Want to show that f is surjective. It suffices to prove that $f_{\mathfrak{m}} : R_{\mathfrak{m}} \longrightarrow (\cap_{\mathfrak{m}\in MaxR} R_{\mathfrak{m}})_{\mathfrak{m}}$ is surjective for each maximal ideal $\mathfrak{m}$. □

Proposition 3.6. Assume $\frac{a}{1} \in S^{-1}R$ and $\frac{a^n}{1} = 0$ for some n. Then there exists $s \in S$ such that $a^n s = 0$ and hence $(as)^n = 0$. While R is reduced, we know $as = 0$ so that $\frac{a}{1} = 0$ in $S^{-1}R$. Conclude that $S^{-1}R$ is reduced. □

Corollary 3.1. By Proposition 3.6, we know $R_{\mathfrak{p}}$ is reduced. While $\mathfrak{p}$ is minimal, the unique prime ideal of $R_{\mathfrak{p}}$ is then 0. Conclude that $R_{\mathfrak{p}}$ is a field. □

8.3.3 Modules

Proposition 3.7. See in the Atiyah. □

Proposition 3.8. For every nonzero $m \in M$, consider the annihilator $\operatorname{ann}_R(m)$. Then $M/Rm \cong R/\operatorname{ann}_R(m)$. By Zorn's Lemma, we can take maximal element $\operatorname{ann}_R(m_0)$ in $\Sigma := \{\operatorname{ann}_R(m) \big| m \neq 0 \in M\}$. And it is clear that this element is a prime ideal in R with $M/Rm_0 \cong \operatorname{ann}_R(m_0)$. Set $M_1 = Rm_0$. Repeating this process, we get a filtration $M = M_0 \supset M_1 \supset \cdots$. While a finite module over noetherian ring is noetherian, hence the filtration is stationary and we get our desired filtration. □

Proposition 3.2. By Proposition 3.8, there is a filtration

$$0 = I_0 \subset I_1 \subset \cdots \subset I_n = R \tag{6}$$

such that $I_i/I_{i-1} \cong A/\mathfrak{p}_i$ for some prime ideal $\mathfrak{p}_i$. Assume $\mathfrak{q}$ appears m times. Note that $0 \to I_{i-1} \to I_i \to R/\mathfrak{p}_i \to 0$ is exact, tensoring by $R_{\mathfrak{q}}$, we get exact sequences

$$0 \longrightarrow I_{i-1} \otimes_R R_{\mathfrak{q}} \longrightarrow I_i \otimes_R R_{\mathfrak{q}} \longrightarrow R/\mathfrak{p}_i \otimes_R R_{\mathfrak{q}} \longrightarrow 0 \tag{7}$$

While for any $\mathfrak{p}_i \neq \mathfrak{q}$, $R/\mathfrak{q}_i \otimes_R R_{\mathfrak{q}} = 0$ and $R/\mathfrak{q} \otimes_R R_{\mathfrak{q}} = \operatorname{Frac}(R/\mathfrak{q})$. Summing on length over $R_{\mathfrak{q}}$, we get $m = \operatorname{length} R_{\mathfrak{q}}$. □

8.3.4 Algebras

⋆ Finiteness

Proposition 3.9. i don't know □

⋆ Integrality

Proposition 3.10. Since L/K is algebraic, for all $\alpha \in L$, there is a minimal polynomial $x^n + \frac{a_1}{a_1'} x^{n-1} + \cdots + \frac{a_n}{a_n'}$, where all a_i and a_i' are in A. Take the multiplication a of all dominators, then $a\alpha$ is a root of $x^n + \frac{a_1 a}{a_1'} x^{n-1} + \cdots \frac{a_n a^n}{a_n'}$ with all coefficients in A. By definition, we know $a\alpha \in B$ and hence $\alpha \in \operatorname{Frac} B$. In conclusion, $\operatorname{Frac} B = L$. □

Proposition 3.11. By Proposition 3.10, we only need to show that for all $b \in B$, $b^{-1} \in B \otimes_A K$. In fact, $B \otimes_A K = S^{-1}B$ where $S = A \setminus \{0\}$. And for all $b \in B$, there is a minimal polynomial $x^n + a_1 x^{n-1} + \cdots + a_n$ with coefficients in A. By definition, we know $a_n \neq 0$. Denote $c = b^{n-1} + a_1 b^{n-2} + \cdots + a_{n-1}$, we get that $bc = -a_n \in A$ so that $b^{-1} = \frac{c}{-a_n}$. □

Proposition 3.12. When B is a field, since A is a subring of B, A is also a field. Conversely, assume A is a field. As B is integral over A, for all nonzero $b \in B$, there is a monic polynomial $x^n + a_1 x^{n-1} + \cdots + a_n$ of b over A with $a_n \neq 0$. Then $b(b^{n-1} + a_1 b_{n-2} + \cdots + a_{n-1}) \in A^{\times}$ so that b is invertible. Conclude that B is a field. □

⋆ Flatness

Proposition 3.13. "$\Rightarrow$ ": Since B is flat over A, then $B \otimes_A A_{\mathfrak{p}}$ is flat over $A_{\mathfrak{p}}$. Note that $A_{\mathfrak{p}} \to B_{\mathfrak{P}}$ factors through $B \otimes_A A_{\mathfrak{p}}$ and $B_{\mathfrak{P}}$ is localization of $B \otimes_A A_{\mathfrak{p}}$. Hence $B_{\mathfrak{P}}$ is flat over $A_{\mathfrak{p}}$.

"$\Leftarrow$ ": Given a short exact sequence $0 \to M' \to M \to M'' \to 0$. Suffices to show $0 \to M' \otimes_A B_{\mathfrak{P}} \to M \otimes_A B_{\mathfrak{P}} \to M'' \times_A B_{\mathfrak{P}} \to 0$ exact for all $\mathfrak{P} \in \operatorname{Spec} B$. While $M \otimes_A B_{\mathfrak{P}} \cong M \otimes_A A_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} B_{\mathfrak{P}}$, done! □

Proposition 3.14. i don't know. □

Proposition 3.15. See in the website, Stacks Project 00HR. □

⋆ Faithful Flatness

Proposition 3.16. Suffice to show that for any A-module M, if $M \otimes_A B = 0$, then $M = 0$. Assume that there is a nonzero element $m \in M$, consider $\operatorname{ann}(m)$. Then $Am \cong A/\operatorname{ann}(m)$ is a nonzero submodule of M. As B is flat over A, we have $A/\operatorname{ann}(m) \otimes_A B \cong B/\operatorname{ann}(m)B$ is a submodule of $M \otimes_A B$. While there exists prime ideal $\mathfrak{p} \in \operatorname{Spec} A$ containing $\operatorname{ann}(m)$ and there is a prime ideal $\mathfrak{P} \in \operatorname{Spec} B$ lying over $\mathfrak{p}$, we have $\operatorname{ann}(m)B \subseteq \mathfrak{P} \neq B$. Thus $B/\operatorname{ann}(m)B \neq 0$, contradiction! □

Corollary 3.3. Note that $IB = I \otimes_A B$. This immediately comes from Proposition 3.16. □

Corollary 3.4. Consider kernel and cokernel of φ. There is an exact sequence

$$0 \longrightarrow \ker \varphi \longrightarrow \mathcal{F} \xrightarrow{\varphi} \mathcal{G} \longrightarrow \operatorname{coker} \varphi \longrightarrow 0 \tag{8}$$

Tensoring by B, we get an exact sequence

$$0 \longrightarrow \ker \varphi \otimes_A B \longrightarrow \mathcal{F} \otimes_A B \xrightarrow{\varphi \otimes_A B} \mathcal{G} \otimes_A B \longrightarrow \operatorname{coker} \varphi \otimes_A B \longrightarrow 0 \tag{9}$$

where by assumption $\varphi \otimes_A B$ is an isomorphism. By Proposition 3.16, we get $\ker \varphi = 0$ and $\operatorname{coker} \varphi = 0$, done! □

8.3.5 Dimension theory

Proposition 3.17. For any $x \neq 0 \in R$, consider the descending chain of ideals

$$(x) \supseteq (x^2) \supseteq \cdots \supseteq (x^n) \supseteq \cdots \tag{10}$$

By Artinian property, we can get that x is invertible. □

8.3.6 Local Rings

Proposition 3.18. Assume that $\mathfrak{m}/\mathfrak{m}^2$ is spanned by $\overline{a_1}, \cdots, \overline{a_n}$ for $a_i \in R$. Get $\mathfrak{m} = (a_1, \cdots, a_n) + \mathfrak{m}^2$. By Nakayama's Lemma, we have $\mathfrak{m} = (a_1, \cdots, a_n)$. □

Corollary 3.5. Immediately comes from Proposition 3.18. □

Proposition 3.19. (1) ⇒ (2) ⇒ (3) ⇒ (4) is clear. Suffices to prove (4) ⇒ (1). As M is finitely generated, by Nakayama's Lemma, we can pick representative elements of a basis of $M/\mathfrak{m}M$ over k, which also generate M. And these elements induces a surjection $R^{\oplus n} \twoheadrightarrow M$ with kernel N. Since $\mathrm{Tor}_1^R(M, k) = 0$, the following sequence is exact

$$0 \longrightarrow N \otimes_R k \longrightarrow R^{\oplus n} \otimes_R k \longrightarrow M \otimes_R k \longrightarrow 0 \tag{11}$$

Note that by our choice, $R^{\oplus n} \otimes_R k \to M \otimes_R k$ is an isomorphism. Hence $N \otimes_R k = 0$ so that by Nakayama's Lemma again, $N = 0$. Conclude that $M \cong R^{\oplus n}$ is free. □

Corollary 3.6. Immediately comes from Proposition 3.19 and the fact that for a finite module over a PID is free if and only if it is torsion-free. □

⋆ Valuation Rings

Proposition 3.20. We may assume $v(a) < v(b)$ so that $v(a + b) \geq v(a)$. Again by the ultrametric inequality, $v(a) \geq \min\{v(a + b), v(-b)\}$. Since $v(-b) = v(b) > v(a)$, we get $v(a) \geq v(a + b)$, done! □

⋆ Henselian Local Rings

Proposition 3.21. Refer to Stacks Project 04GG. □

Corollary 3.7. Obviously □

Proposition 3.22. For any $\frac{a}{b} \in \mathrm{Frac}(R) \backslash R$ integral over R, assume that $\gcd(a, b) = 1$. There exists a monic polynomial $x^n + a_1 x^{n-1} + \cdots + a_n$ where $a_i \in R$ s.t. $(\frac{a}{b})^n + a_1(\frac{a}{b})^{n-1} + \cdots + a_n = 0$. Thus $a^n = -b(a_1 a^n + \cdots + a_n b^{n-1})$, get $b \mid a^n$ which contradicts to our assumption, $\gcd(a, b) = 1$. □

Proposition 3.23. Consider $f : A_{\mathfrak{p}} \longrightarrow B_{\mathfrak{p}}$. Then the maximal ideal of $B_{\mathfrak{p}}$ lies over $\mathfrak{p}$. □

Proposition 3.24. By definition. □

Proposition 3.25. For any minimal prime ideal $\mathfrak{p}$ of A containing kernel, consider set $V = V(\varphi(\mathfrak{p}))$ in $\operatorname{Spec} B$, which by assumption is nonempty. As A, B are both noetherian, V is finite. Suppose that preimages of all $\mathfrak{P} \in V(\varphi(\mathfrak{p}))$ are not $\mathfrak{p}$, then each $\mathfrak{P}_i$ gives an $a_i \notin \mathfrak{p}$. Set $a = \prod_i a_i$, then $a \notin \mathfrak{p}$ and $\varphi(a) \in \cap_i \mathfrak{P}_i$ so that $\varphi(a) \in \sqrt{\varphi(\mathfrak{p})}$. Note that $\varphi^{-1}(\varphi(\mathfrak{p})) = \mathfrak{p}$, we get $a \in \mathfrak{p}$, contradiciton. $\square$

Corollary 3.8. Consider ring homomorphism $\widetilde{\varphi} : A/\mathfrak{p} \to B/\varphi(\mathfrak{p})B$. Then apply Proposition 3.25, done! $\square$

8.4 Linear Algebra 线性代数

Proposition 4.1. Consider the maximal linearly independent system $\square$

8.5 Representation Theory 表示论

Theorem 5.1. See in the note. $\square$

Theorem 5.2. Note that $\chi^{g_i} = \chi$, we have

$$\begin{aligned} \langle Res_N^G(\chi), \varphi^{g_i} \rangle &= \langle Res_N^G(\chi^{g_i}), \varphi^{g_i} \rangle \\ &= \langle Res_N^G(\chi)^{g_i}, \varphi^{g_i} \rangle \\ &= \langle Res_N^G(\chi), \varphi \rangle \\ &= e \end{aligned}$$

For the proof of showing φ^{g_i} are all the direct summand of $Res_N^G(\chi)$, see in the book Representation Theory of Finite Group Extensions. $\square$

8.6 Algebraic Geometry

8.6.1 Classic examples

Proposition 6.1. See in the website, invertible sheaves of the projective line. $\square$

8.6.2 Spectrum

Proposition 6.2. It is clear that for all $\mathfrak{p} \in \operatorname{Spec} B$, $\ker \varphi \subseteq \varphi^{-1}(\mathfrak{p})$ so that $f(\mathfrak{p}) \in V(\ker \varphi)$. Thus $\operatorname{im} f \subseteq V(\ker \varphi)$ $\square$

Corollary 6.1. Immediately comes from Proposition 6.2 and Corollary 3.8. $\square$

Corollary 6.2. Immediately comes from Proposition 6.2. $\square$

Proposition 6.3. For different $\mathfrak{p}_1$ and $\mathfrak{p}_2$ in $\operatorname{Spec} B$, there must exists $a \in \mathfrak{p}_1 \setminus \mathfrak{p}_2$ so that $\varphi^{-1}(a) \subseteq \varphi^{-1}(\mathfrak{p}_1) \setminus \varphi^{-1}(\mathfrak{p}_2)$ and hence $f(\mathfrak{p}_1) \neq f(\mathfrak{p}_2)$. $\square$

Proposition 6.4. $\square$

8.6.3 Properties of morphisms

⋆ Separatedness

Proposition 6.5. Since the problem is locally, we may assume $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$ are both affine. Consider the following commutative diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "g"] & Y \arrow[d] \\ Y \arrow[r] & \operatorname{Spec} \mathbb{Z} \end{tikzcd} \tag{12}$$

By universal property of fibered product, we get a lifting $(f, g) : X \to Y \times Y$. Note that the set $\{x \in X \big| f(x) = g(x)\}$ is the preimage of Δ_Y under (f, g), which is a closed subset of X since Y is separated. As f and g agree on an open dense subset U of X, we get $U \subset \{x \in X \big| f(x) = g(x)\}$ and hence $\{x \in X \big| f(x) = g(x)\} = X$.

Now we have two ring homomorphisms $f^\sharp : B \to A$ and $g^\sharp : B \to A$ satisfying that for all $\mathfrak{p} \in \operatorname{Spec} A$, $f(\mathfrak{p}) = g(\mathfrak{p})$. Remains to show $f^\sharp = g^\sharp$. As X is reduced, it suffices to show that for all $b \in B$, $V(f^\sharp(b) - g^\sharp(b)) = X$. Since U is dense in X, it suffices to show that $U \subseteq V(f^\sharp(b) - g^\sharp(b))$.

In fact, since for all $\mathfrak{p} \in U$, we can take an affine open neighbourhood $D(s) \subseteq U$. Then $B \xrightarrow{f^\sharp} A \to A_s$ and $B \xrightarrow{g^\sharp} A_s$ are same. Hence we get $\frac{f^\sharp(b)}{1} = \frac{g^\sharp(b)}{1}$ in A_s and $(f^\sharp(b) - g^\sharp(b))s^n = 0$ for some n. As $s \notin \mathfrak{p}$, $f^\sharp(b) - g^\sharp(b) \in \mathfrak{p}$ and we are done! □

⋆ Flatness

Proposition 6.6. "$\Leftarrow$ ": for all affine open subset $V = \operatorname{Spec} A$ of Y, and for all affine open subset $U = \operatorname{Spec} B$ of $f^{-1}(V)$, given a short exact sequence of A-modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0 \tag{13}$$

Tensoring with $A_\mathfrak{p}$ over A for all $\mathfrak{p} \in \operatorname{Spec} A$, we get a short exact sequence

$$0 \longrightarrow M'_\mathfrak{p} \longrightarrow M_\mathfrak{p} \longrightarrow M''_\mathfrak{p} \longrightarrow 0 \tag{14}$$

By assumption, $B_\mathfrak{P}$ is flat over $A_\mathfrak{p}$ for all $\mathfrak{P} \in \operatorname{Spec} B$ lying over $\mathfrak{p}$, we get a short exact sequence

$$0 \longrightarrow M'_\mathfrak{p} \otimes_{A_\mathfrak{p}} B_\mathfrak{P} \longrightarrow M_\mathfrak{p} \otimes_{A_\mathfrak{p}} B_\mathfrak{P} \longrightarrow M''_\mathfrak{p} \otimes_{A_\mathfrak{p}} B_\mathfrak{P} \longrightarrow 0 \tag{15}$$

Note that $M_\mathfrak{p} \otimes_{A_\mathfrak{p}} B_\mathfrak{P} = M \otimes_A B \otimes_B B_\mathfrak{P}$. Hence by local property of module homomorphism, we get a short exact sequence

$$0 \longrightarrow M' \otimes_A B \longrightarrow M \otimes_A B \longrightarrow M'' \otimes_A B \longrightarrow 0 \tag{16}$$

Conclude that B is flat over A so that f is flat.

"$\Rightarrow$ ": for any $x \in X$, consider the following commutative diagram

$$\begin{tikzcd} \operatorname{Spec} \mathcal{O}_{X,x} \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} \mathcal{O}_{Y,f(x)} \arrow[r] & Y \end{tikzcd} \tag{17}$$

By base change, we get a morphism $\operatorname{Spec}\mathcal{O}_{X,x} \to \operatorname{Spec}\mathcal{O}_{Y,f(x)}$. By assumption, this morphism is flat and hence $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$. □

Proposition 6.7. For a generic point η' of X, assume $f(\eta') = y$. Take an affine open neighbourhood U of y and affine open neighbourhood $W \subseteq f^{-1}(U)$ of η'. Assume $U = \operatorname{Spec} A$ and $W = \operatorname{Spec} B$. Then B is flat over A. By Going-down Theorem, there exists some generalization x of η' mapping to η. While η' is generic point, this forces $y = \eta$. By assumption, the fiber at η is irreducible. This implies that X has unique generic point and hence irreducible. □

⋆ Faithful Flatness

Proposition 6.8. Since the problem is local on the target, we may assume $Y = \operatorname{Spec} A$ for some ring A. Assume $\mathcal{F}$ and $\mathcal{G}$ are given by A-modules M and N respectively. Now suffices to show for all $y \in Y$, φ_y is an isomorphism.

As f is surjective, there is $x \in X$ such that $f(x) = y$. In addition, as f is flat, $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,y}$. Then by Proposition 3.15, $\mathcal{O}_{X,x}$ is faithfully flat over $\mathcal{O}_{Y,y}$. While $M_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x} \to N_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x}$ is isomorphic by assumption, we get $M_y \to N_y$ is isomorphic by Corollary 3.4. □

Corollary 6.3. Since S and T are affine over Y, we may assume $Y = \operatorname{Spec} A$, $S = \operatorname{Spec} B$ and $T = \operatorname{Spec} C$ for some $A, B \in \mathbf{Alg}_C$. Note that $\mathcal{O}_S$ and $\mathcal{O}_T$ can be identified with the quasi-coherent sheaves associated to B and C respectively. Then the morphism $S \to T$ corresponds to a morphism of A-algebras $C \to B$. By Proposition 6.8, we get $C \to B$ is an isomorphism and hence $S \to T$ is an isomorphism. □

8.6.4 Dimension Theory

Proposition 6.9. See in the website, dimension of irreducible affine variety is same as any open subset. □

Proposition 6.10. Since constructible can be written as finite union of locally closed subsets, assume that $Y = \cap_{i=1}^{n} U_i \cap V_i$. In particular, by taking irreducible components of each V_i, we may assume that V_i are irreducible. Then $\dim \overline{Y} \geq \dim Y \geq \max_i\{\dim(U_i \cap V_i)\}$. Note that $\overline{Y} = \cup_{i=1}^{n} \overline{U_i \cap V_i}$ and by Proposition 6.9, we immediately get

$$\dim \overline{Y} = \max_i\{\dim(\overline{U_i \cap V_i})\} = \max_i\{\dim V_i\} = \max_i\{\dim(U_i \cap V_i)\} \tag{18}$$

Thus $\dim Y = \dim \overline{Y}$. □

Proposition 6.11. Since $\dim(X \times_k K) = \dim X$, suffices to check of pure dimension. While purely dimensional property can be checked locally, we may assume $X = \operatorname{Spec} A$ for A finitely generated k-algebra. As reduction does not change topology, we may assume A is integral domain, denote $F := \operatorname{Frac}(A)$. Then by Theorem 1.8A in Hartshorne Chapter I, we know $\dim A = \operatorname{trdeg}(F/k)$. Assume $\dim A = d$ and $x_1, \cdots, x_d \in F$ transcandence basis of F over k. Denote $E := k(x_1, \cdots, x_d)$, then F is algebraic over E.

Note that $X \times_k K$ is flat over X, by Going-down Theorem for flat morphism, generic points of $X \times_k K$ would all map to $(0) \in X$. Now for every minimal prime ideal $\mathfrak{p}$ of $A \otimes_k K$, it would corresponds to a minimal prime ideal $\mathfrak{P}$ of $F \otimes_k K$ and again by Theorem 1.8A, we have

$$\dim \overline{\{\mathfrak{p}\}} = \operatorname{trdeg}(\operatorname{Frac}(A \otimes_k K/\mathfrak{p})/K) = \operatorname{trdeg}(\operatorname{Frac}(F \otimes_k K/\mathfrak{P})/K) \tag{19}$$

Denote $L := \operatorname{Frac}(F \otimes_k K/\mathfrak{P})$. Note that $F \otimes_k K$ is integral over $E \otimes_k K$ and the latter one is isomorphic to a localization of $K[x_1, \cdots, x_d]$ which is an integral domain. Also as $F \otimes_k K$ is flat over $E \otimes_k K$, we get $\mathfrak{P}$ contracts to (0) in $E \otimes_k K$. Hence there is an injection

$$E \otimes_k K \hookrightarrow F \otimes_k K \twoheadrightarrow F \otimes_k K/\mathfrak{P} \hookrightarrow L \tag{20}$$

Hence $\operatorname{trdeg}(L/K) = \operatorname{trdeg}(\operatorname{Frac}(E \otimes_k K)/K) = \operatorname{trdeg}(K(x_1, \cdots, x_d)/K) = d$, done! $\square$

8.6.5 Universal injectivity

Proposition 6.12. I don't know $\square$

Proposition 6.13. As composition of projection map and $\Delta_{X/Y}$ is identity map of X, if f is universally injective, then projection map is injective and hence $\Delta_{X/Y}$ is surjective.

Conversely, for any morphism $Z \to Y$, it suffices to show that $\Delta_{X \times_Y Z/Z}$ is surjective. Note that there is a fibered diagram

$$\begin{tikzcd} X \times_Y Z \arrow[r, "\Delta_{X\times_Y Z/Z}"] \arrow[d] & X \times_Y Z \times_Z X \times_Y Z \cong X \times_Y X \times_Y Z \arrow[d] \\ X \arrow[r, "\Delta_{X/Y}"] & X \times_Y X \end{tikzcd} \tag{21}$$

Then as base change preserves surjectivity, we get $\Delta_{X \times_Y Z/Z}$ is surjective. $\square$

8.6.6 Quasi-coherent sheaves

Proposition 6.14. By Proposition 2.1, closedness is local. Hence we can reduce to affine case. Assume $X = \operatorname{Spec} A$ and $\mathcal{F} = \widetilde{M}$ for some finite A-module M. Pick generators $m_1, \cdots, m_n$ of M. Then $M_x = 0$ if and only if $(m_i)_x = 0$ for all i. Conclude that $\operatorname{Supp} \mathcal{F} = \cup_{i=1}^n \operatorname{Supp} m_i$ is closed. $\square$

8.6.7 Vector bundles

Proposition 6.15. Assume $x_0, \ldots, x_n$ are the homogeneous coordinates of $\mathbb{P}_k^n$. Then transition functions of $\mathcal{O}_{\mathbb{P}_k^n}(-1)$ are given by $(D_+(x_i), \frac{x_i}{x_j})$. Define map $\mathcal{O}_{\mathbb{P}_k^n}(-1) \to \mathcal{O}_{\mathbb{P}_k^n}^{\oplus n+1}$ by sending s to $(\frac{x_0}{x_i}s, \ldots, \frac{x_n}{x_i}s)$ on $D_+(x_i)$. Then it is easy to check that these local maps glue together and give a global morphism $\mathcal{O}_{\mathbb{P}_k^n}(-1) \to \mathcal{O}_{\mathbb{P}_k^n}^{\oplus n+1}$. $\square$

Corollary 6.4. Take affine cover $\{U_i\}$ of X such that $\mathcal{E}\big|_{U_i}$ is trivial. Then we have a canonical inclusion $\mathcal{O}_{\mathbb{P}_X(\mathcal{E})}(-1)\big|_{\pi^{-1}(U_i)} \to \pi^*\mathcal{E}\big|_{\pi^{-1}(U_i)}$ by Proposition 6.15. These local maps glue together and give a global morphism $\mathcal{O}_{\mathbb{P}_X(\mathcal{E})}(-1) \to \pi^*\mathcal{E}$. $\square$

Proposition 6.16. "$\Rightarrow$ ": Assume $\mathbb{P}(\mathcal{E})$ is irreducible. Note that the natural morphism $\pi : \mathbb{P}(\mathcal{E}) \to X$ is surjective. We know $X = \pi(\mathbb{P}(\mathcal{E}))$ is irreducible.

"$\Leftarrow$ ": Assume X is irreducible with generic point η. Then the fiber of π at η is a projective space over $k(\eta)$ and hence irreducible. Also, affine locally, $\mathbb{P}^N_A \to \operatorname{Spec} A$ is smooth so that π is smooth. By Proposition 6.7, we get $\mathbb{P}(\mathcal{E})$ is irreducible. □

8.6.8 Exceptional locus

Proposition 6.17. Obviously, $\{x \in X | g \text{ is not defined at } f(x)\} \subseteq X \setminus U_X$. We only need to show the other containment. Assume g is defined at $f(x)$. Then by definition of rational map, there are open neighbourhoods $U \ni x$ and $V \ni x$ such that $f(U) \subseteq V$ and g is defined on V.

As U_X is open dense subset of X, $U_X \cap U$ is nonempty and dense in U. Hence $g \circ f|_U$ and $U \hookrightarrow X$ agree on $U_X \cap U$. By Proposition 6.5, $g \circ f|_U$ is just the open immersion.

Hence we can replace V by $V \cap g^{-1}(U)$, which is still an open neighbourhood of $f(x)$. Applying same argument for $f \circ g|_V$, we conclude that $U \subseteq U_X$ and $V \subseteq U_Y$, done! □

Proposition 6.18. As D is not contained in $\operatorname{Sing} Y$, Y is regular at η so that $\mathcal{O}_{Y,\eta}$ is a DVR. Assume Z is an irreducible component of Y such that $D \subseteq Z$ and ξ is the generic point of Z, then $\mathcal{O}_{Y,\xi} = \operatorname{Frac}(\mathcal{O}_{Y,\eta})$.

Since U_Y is dense in Y, $\xi \in U_Y$ so that g induces a map $\operatorname{Spec} \mathcal{O}_{Y,\xi} \to U_X$. And we have a commutative diagram

$$\begin{tikzcd}
\operatorname{Spec} \mathcal{O}_{Y,\xi} \arrow[r] \arrow[d] & U_X \arrow[d, "f"] \\
\operatorname{Spec} \mathcal{O}_{Y,\eta} \arrow[r] \arrow[ur, dashed] & Y
\end{tikzcd} \tag{22}$$

Applying valuative criterion of properness, there exists unique lifting $\operatorname{Spec} \mathcal{O}_{Y,\eta} \to X$. Hence there exists $\delta \in U_X$ such that $f(\delta) = \eta$. By definition, we know $\eta \notin \operatorname{Ex}(g)$. □

8.6.9 Ramification theory

Proposition 6.19. As f is a finite morphism of degree d, the stalk $(f_*\mathcal{O}_X)_Q$ is a free $\mathcal{O}_{Y,P}$-module of rank d. Tensoring by $k(Q)$, we get a d-dimensional vector space over $k(Q)$. Note that by Proposition 3.9, we have that

$$\begin{aligned}
(f_*\mathcal{O}_X)_Q \otimes_{\mathcal{O}_{Y,Q}} k(Q) &\cong (\oplus_{i=1}^r \mathcal{O}_{X,P_i}) \otimes_{\mathcal{O}_{Y,Q}} k(Q) \\
&\cong \oplus_{i=1}^r \mathcal{O}_{X,P_i} / \mathfrak{m}_Q \mathcal{O}_{X,P_i} \\
&\cong \oplus_{i=1}^r \mathcal{O}_{X,P_i} / \mathfrak{m}_{P_i}^{e_{P_i}}
\end{aligned} \tag{23}$$

Counting dimension, we get

$$d = \sum_{i=1}^r e_{P_i} \tag{24}$$

□

8.6.10 Intersection theory

Proposition 6.20. First note that $p^*(Z \cdot D^d) = p^*Z \cdot (p^*D)^d$ in Chow group. Suffice to show for all closed point $x \in X$ and $f^*[x] = \sum_i m_i[y_i]$, we have that $\chi(X, x) = \sum_i m_i \chi(X_K, y_i)$. In fact, this immediately comes from the equation

$$[k(x) : k] = \sum_i \mathrm{length}(\mathcal{O}_{X_K \times_X k(x), y_i}) \cdot [k(y_i) : K] \tag{25}$$

□

8.7 Homological Algebra 同调代数

Proposition 7.1. Here we only prove for the injective version. As $I^\bullet$ is acyclic, denote $\ker \partial_n = Z_n$. Then there are short exact sequences

$$0 \longrightarrow Z_n \longrightarrow I^n \longrightarrow Z_{n+1} \longrightarrow 0 \tag{26}$$

Since $I^\bullet$ is bounded-below, we may assume for all negative i, $I^i = 0$. Hence $Z_2 = I^1$. Consider the short exact sequence

$$0 \longrightarrow I^1 \longrightarrow I^2 \longrightarrow Z_3 \longrightarrow 0 \tag{27}$$

As I^1 is injective, the exact sequence splits so that Z_3 is a summand of I^2 and hence injective. By induction, we get that all those short exact sequences split. Now we have projections $p_n : I^n \to Z_n$ and inclusions $s_{n+1} : Z_{n+1} \to I^n$. Consider $D_n := s_n \circ p_n : I^n \to I^{n-1}$. In fact we have that

$$\partial \circ D_n + D_{n+1} \circ \partial = p_n + s_{n+1} = \mathrm{id}_{I^n} \tag{28}$$

Hence $I^\bullet$ is contractible. □